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Comparative Pan-Cancer Analysis Reveals Pathway Signatures Distinguishing Lung Adenocarcinoma from Squamous Cell Carcinoma

Data Science Project SoSe 2024

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Heidelberg, 08.07.2024
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Lung adenocarcinoma (LUAD) is a predominant form of lung cancer characterized by poor survival rates due to late-stage diagnoses and complex genetic and epigenetic heterogeneity (Siegel *et. al*, 2023; TCGA, 2014). This project aims to elucidate the molecular features of LUAD which drive cancer progression and formation, as well as differentiate LUAD from lung squamous cell carcinoma (LUSC). Utilizing RNA sequencing data from The Cancer Genome Atlas (TCGA), shortened to protein encoding genes, we performed Gene Set Enrichment Analysis (GSEA) and Gene Set Variation Analysis (GSVA) on LUAD patient samples and pan-caner data. These analyses revealed enrichment in hallmark pathways such as angiogenesis, proliferation, and apoptosis across LUAD samples. However, individual patients exhibited unique oncogenic signature enrichments without distinct clustering when visualized using Uniform Manifold Approximation and Projection (UMAP), indicating significant heterogeneity, which might arise due to several molecular mechanisms.

In contrast, our GSVA of pan-cancer data demonstrated distinct clustering of different cancer types, highlighting pathway-specific regulatory patterns. Further, logistic regression analysis identified 13 key pathways that effectively distinguish LUAD from LUSC, achieving an accuracy of 92.6% and an area under the curve (AUC) of 0.9779. This analysis underscores the need for personalized therapeutic approaches in LUAD treatment due to the diverse genetic, epigenetic, and microenvironmental factors influencing tumor biology. Overall, our findings contribute to a deeper understanding of LUAD's molecular landscape and supports the development of targeted diagnostic and treatment strategies.

Abbreviations

ACC Adrenocortical carcinoma

AIC Akaike Information Criterion

AUC Area under the curve

BLCA Bladder urothelial carcinoma

BRCA Breast invasive carcinoma

CESC Cervical squamous cell carcinoma and endocervical adenocarcinoma

CHOL Cholangiocarcinoma

COAD Colon adenocarcinoma

DLBC Lymphoid neoplasm diffuse large b-cell lymphoma

ES Enrichment score

ESCA Esophageal carcinoma

FC Fold change

GBM Glioblastoma multiforme

GSEA Gene set enrichment analysis

GSVA Gene set variation analysis

KICH Kidney Chromophobe

KIRC Kidney renal clear cell carcinoma

KIRP Kidney renal papillary cell carcinoma

LAML Acute myeloid leukemia

LGG brain lower grade glioma

LIHC Liver hepatocellular carcinoma

LUAD Lung adenocarcinoma

LUSC Lung squamous cell carcinoma

MESO Mesothelioma

MPC model predictive control

MSigDB Molecular signatures database

NES Normalized enrichment score

OV Ovarian serous cystadenocarcinoma

PAAD Pancreatic adenocarcinom

PCA Principal component analysis

PCPG Pheochromocytoma and Paraganglioma

PRAD Prostate adenocarcinoma

READ Rectum adenocarcinoma

ROC Receiver operator characteristic

SARC Sarcoma

SKCM Skin cutaneous melanoma

STAD Stomach adenocarcinoma

TCGA The Cancer Genome Atlas

TGCT Testicular germ cell tumors

THCA Thyroid carcinoma

THYM Thymoma

TPM Transcripts per million

UCEC Uterine corpus endometrial carcinoma

UCS Uterine carcinosarcoma

UMAP Uniform manifold approximation and projection

UVM Uveal melanoma

1 Introduction

1.1 Cancer

Cancer is a disease characterized by the uncontrolled growth and spread of cells from various origins due to damage in critical genetic programs. These genetic alterations lead to the development of specific biological features, known as cancer hallmarks. The hallmarks of cancer include capabilities such as sustaining proliferative signaling and genome instability (Hanahan and Weinberg, 2000). Recent discoveries have expanded these hallmarks to include properties resulting from the tumor microenvironment, which consists of both cancerous and ostensibly healthy cells, such as macrophages. Today the new and established hallmarks are consolidated in the updated list of cancer hallmarks (Hanahan, 2022).
Cancer is the second leading cause of death worldwide, accounting for 10 million deaths annually, with 1.9 million new cases emerging each year in the US alone (a trend that is increasing) (Siegel *et al.*, 2023). Thus, identifying and characterizing new targets for cancer detection and treatment is crucial to reducing its lethality.

1.2 LUAD

LUAD is a type of non-small-cell lung cancer and yet remains the most common primary lung cancer, accounting for 40% of lung cancers. Due to late diagnoses in middle and late stages, LUAD has a 5-year survival rate of only 12-15%. This low survival rate, combined with the high number of cases, makes LUAD the leading cause of cancer-related deaths in the US, with 158,000 deaths in 2015.
LUAD is often associated with perturbations and deregulation of specific pathways and genes, such as the mitogen-activated protein kinase (MAPK) pathway (involved in cell survival and proliferation), the tumor suppressor gene P53, and the epithelial growth factor receptor (EGFR), among others (Ding *et al.*, 2008). Identifying these perturbations and dysregulations is essential for diagnosing LUAD. Moreover, developing and improving therapy options are key to reducing deaths caused by LUAD (Siegel *et al.*, 2023).

1.3 LUAD vs. LUSC

To effectively tackle cancer and choose the right therapeutic approach for each patient, it is essential to fully differentiate between closely related cancer types, such as LUAD and LUSC. Historically, these two tumor types were categorized by their histology and grouped together as non-small-cell lung cancer (NSCLC). However, recent studies suggest that differentiating between the two and providing tailored treatment can increase therapeutic success. Therefore, it is crucial to identify and characterize the differences between LUAD and LUSC to distinguish them accurately, for instance, by visualizing and analyzing differences and similarities in expression data (Wang *et al.*, 2022).

2 Material and Methods

2.1 Data

The following analyses are mainly based on three datasets. The first dataset comprises RNA sequencing values for 60,498 genes across 9,741 patients from TCGA (Cooper *et al.*, 2018), covering 33 different tumor types. In addition to these expression data, clinical annotations such as tumor type, gender, or vital status are assigned to the patients.
A focus of this project is the examination of LUAD. For this purpose, a dataset of 58 patients is available, which includes expression data from both cancerous and normal tissues, as well as clinical annotations. All data sets containing expression values have been log2-transformed and normalized using transcripts per million (TPM). A final dataset stores 46 Hallmark gene sets along with the associated genes.

2.1.1 Data cleaning

In order to include only protein-coding genes in both datasets, biotype filtering was performed using the `biomaRt` package (Durinck *et al.*, 2005). This step reduced the gene count from 60,498 to 19,579 in the pan-cancer dataset and from 19,624 to 19,144 in the LUAD data. To further reduce the dimensionality of the first dataframe, low-variance filtering was applied. This involved computing the variance for each gene and subsequently excluding those that fell into the lower p50 quantile.

2.2 Computational Tools

2.2.1 Dimension reduction using PCA and UMAP

The Principal Component Analysis (PCA) (Shlens, 2014) and Uniform Manifold Approximation and Projection (UMAP) methods were used to visualize high-dimensional datasets in two dimensions. UMAP is a dimensionality reduction technique that preserves local and global structures when projecting high-dimensional data into a lower-dimensional space (McInnes *et al.*, 2020). Based on manifold learning, UMAP constructs a high-dimensional graph of the data and optimizes it for a low-dimensional projection, maintaining relationships from the original space. It is scalable to large datasets and offers flexible parameter settings to balance local and global structure preservation. UMAP was specifically used for pan-cancer and LUAD-specific analysis to visualize the data, identify groupings, and examine regulatory patterns within different patients concerning various pathways.

2.2.2 Gene Set Variation Analysis

Gene Set Variation Analysis (GSVA) is an unsupervised method for gene set enrichment, focusing on pathways rather than individual genes. It enables pathway-centered models to identify up- or down-regulated processes in tumor tissues using tumor RNA-seq data alone, without needing data from healthy tissue.
Initially, RNA-seq data is transformed into ranks, making the analysis robust against outliers. The enrichment score, indicating a gene set's relative activity in a sample, is calculated using a Kolmogorov-Smirnov-like statistic. This score reflects the difference between the distribution of ranks for genes in the set and those for the remaining genes. Scores are normalized for comparability across samples.

In this project, we applied GSVA to a pan-cancer dataset and compared LUAD and LUSC using the top 30 gene sets with the highest variance (Hänzelmann *et al.*, 2013).

2.2.3 Gene Set Enrichment Analysis

Gene Set Enrichment Analysis (GSEA)(Subramanian *et al.*, 2005) is a method to interpret RNA expression data, like RNA-seq from TCGA. It uses gene sets—groups of genes with similar functions or regulations—to analyze genome-wide expression data from two classes (e.g., LUAD-tissue vs. healthy-tissue). Genes are ranked based on their expression levels and class differentiation. GSEA checks if genes in a set S are randomly distributed or clustered at either end of the ranked list L. Enriched gene sets indicate up- or downregulation related to class differences.
Key GSEA values include the enrichment score (ES), which shows the degree of gene set enrichment, and the significance (nominal p-value), determined by a permutation test. The ES can be normalized (NES), and the p-value adjusted for multiple tests (Subramanian *et al.*, 2005).

GSEA was used to analyze LUAD patients' gene expression compared to healthy tissue, using hallmark and oncogenic gene sets from the Molecular Signatures Database (MSigDB). The same method analyzed mean fold-change (FC) values between healthy and cancerous tissue, using the `fgsea` package (Korotkevich *et al.*, 2019).

2.2.4 Logistic regression

Logistic regression was applied to predict the probability P(Y=1 l X) for binary outcomes, utilizing independent variables (predictors) that can be continuous or categoricalto. The model is expressed as:

$$\log\left(\frac{p_1}{1-p_1}\right) = \beta_0 + \beta_1 X_1 + ... + \beta_i X_i$$

where: p is the probability of the event, β_0 is the intercept, β_1, β_i are the coefficients for the predictors X_1, X_i (Wilson and Lorenz, 2015). The `glm` function was employed to create this model, predicting whether a sample is from a LUAD or LUSC patient. The top 30 variable pathways identified by GSVA underwent backward selection using the `step` function to determine significant predictors, resulting in 13 pathways. The dataset, containing enrichment scores for these pathways from 515 LUAD and 498 LUSC patients, was devided into training and test sets (80%/20%), with LUAD labelled as "1" and

LUSC as “0”. Evaluation included statistical test and Receiver Operator Characteristic (ROC)-curve analysis to calculate the AUC .

3 Results

3.1 TCGA Pan-cancer analysis

3.1.1 Different tumor types show distinct clustering using GSVA and UMAP

The genetic diversity and heterogeneity of the pan-cancer dataset were analyzed using GSVA. The data was narrowed down to the top 30 genesets with the highest variance for further analysis. Average enrichment scores for each tumor type were calculated to compare the different tumor types, showing recognizable variation in pathway activity between patients. Hierarchical clustering using complete linkage was performed on the heatmap, revealing distinct groupings of pathways and patients (see figure 3.1 A). To visualize the diversity among patients, a PCA was performed on the GSVA enrichment scores for all 73 genesets. Additionally a UMAP was computed on the scores of the PCA. The visualization of the first two UMAP dimensions can be seen in figure 3.1 A. These methods resulted in 11 clusters, 8 of which were visibly isolated from the rest. To identify the basis of the grouping, the clinical annotations of these patients were colored one after another. Most annotations such as age or gender did not cluster according to the UMAP plot. The only clinical annotations that clustered similarly were the tumor types, for example, breast cancer (BRCA, orange on the right) or prostate adenocarcinoma (PRAD, blue on the far right) show distinct groups. Both cancer types show a very different regulation than the others, as seen in figure 3.1 A. In the average of the patients with prostate adenocarcinoma the pathways “HALLMARK_MYC_TARGETS_V1”, “HALLMARK_E2F_TARGETS” and “HALLMARK_G2M_CHECKPOINT” are very down-regulated, but this is also the case in thyroid carcinoma (THCA) and pheochromocytoma and paraganglioma (PCPG). What distinguishes PRAD from the rest of the tumors is the upregulation of “HALLMARK_ANDROGEN_RESPONSE” and could therefore be a possible marker for this tumor type. In the center of the image, it becomes more difficult to separate the tumor types and assign them to specific clusters. If you compare this to the results of the heatmap, tumors that have similar activity for most pathways tend to cluster together. Two tumors that are very alike and make it very difficult to differentiate them based on their very similar pathway activity are colon adenocarcinoma a (COAD) and rectal adenocarcinoma a (READ).

3.1.2 Cancer types show differences and similarities in pathway activity

In addition to clustering by tumor type, patients can be grouped by similarities and differences in different pathway regulations. For example, the activity of the hallmark geneset called “PI3K_AKT_MTOR_signaling” is up-regulated in BRCA and gastrointestinal tumors such as Liver Hepatocellular Carcinoma (LIHC), while it is down-regulated in other patients or tumors, which can be seen in figure 3.2. One pathway that shows a similar scatter in activity levels between tumors relates to oxidative phosphorylation. In LUAD, for example, this is very strongly upregulated compared to other tumors. This was investigated with the help of UMAP plots and staining of the

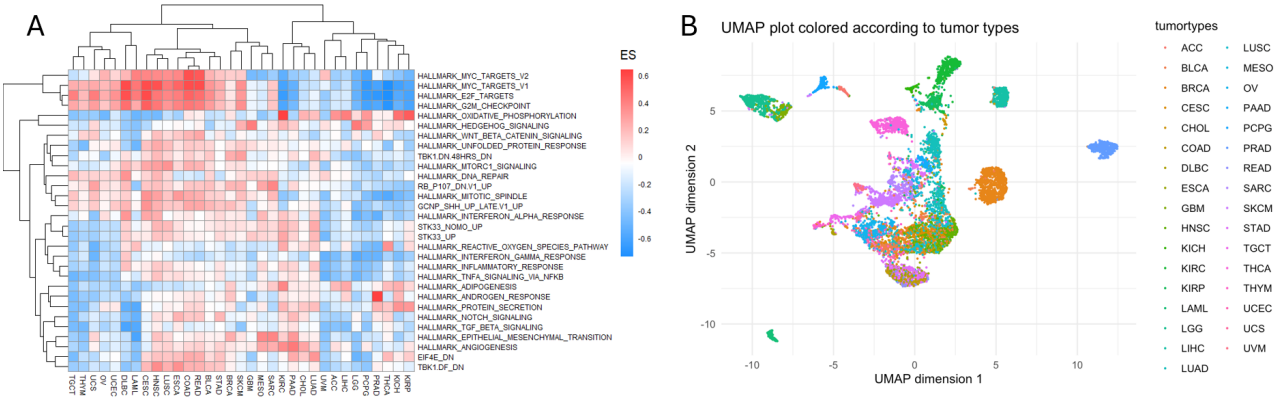

Figure 3.1: Heatmap and UMAP on Pan-Cancer data. (A) Here a heatmap of GSVA with average enrichment scores of patients assigned to each tumor type in columns and top 30 variant gene sets in rows can be seen. (B) This figure shows a UMAP plot of pathway activity data colored by tumor type, indicating specific tumor type clustering.

respective enrichment scores regarding the pathways, therefore the functions `irlba` (Baglama *et al.*, 2022) and `umap` (McInnes *et al.*, 2020) were used.

Furthermore patients with the same tumor type often show similar patterns of activity. However, there are also genesets that show similar expression not only within a tumor but across tumor types. These include gene sets that affect coagulation or Myc.

3.2 Focused analysis

3.2.1 Average LUAD patients show enrichment in certain hallmark pathways

First, a GSEA on the mean FC values for each gene across all LUAD patients, comparing healthy tissue expression data to LUAD tissue expression data, was performed. To do this, a ranked and named list of the mean FC values and a list containing the corresponding gene sets were created, and the `fgsea` function was applied. This analysis revealed significant enrichment in distinct hallmark gene sets associated with proliferation, metastasis, angiogenesis, apoptosis, and genome repair mechanisms. The specific gene sets in which significant enrichment was found are shown in figure 3.3.

3.2.2 LUAD patients show individual enrichment but poor clustering using GSEA and UMAP

After performing GSEA using the `fgsea` function on the FC data of each LUAD patient (LUAD tissue vs. healthy tissue) with both hallmark and oncogenic signature gene sets (C6 from MSigDB), each patient showed an individual enrichment pattern. This was also visualized in a heatmap created from the NES values computed from a GSEA on the oncogenic signature (C6) genesets from the `msigdbr`-package (Dolgalev, 2022), and stored in an enrichment matrix (top 30 most variant) (as seen in figure 3.4). Performing a UMAP on the GSEA data (hallmark- and C6-genesets) revealed that these patients cluster very poorly, showing no distinct subsets as can be seen in the appendix.

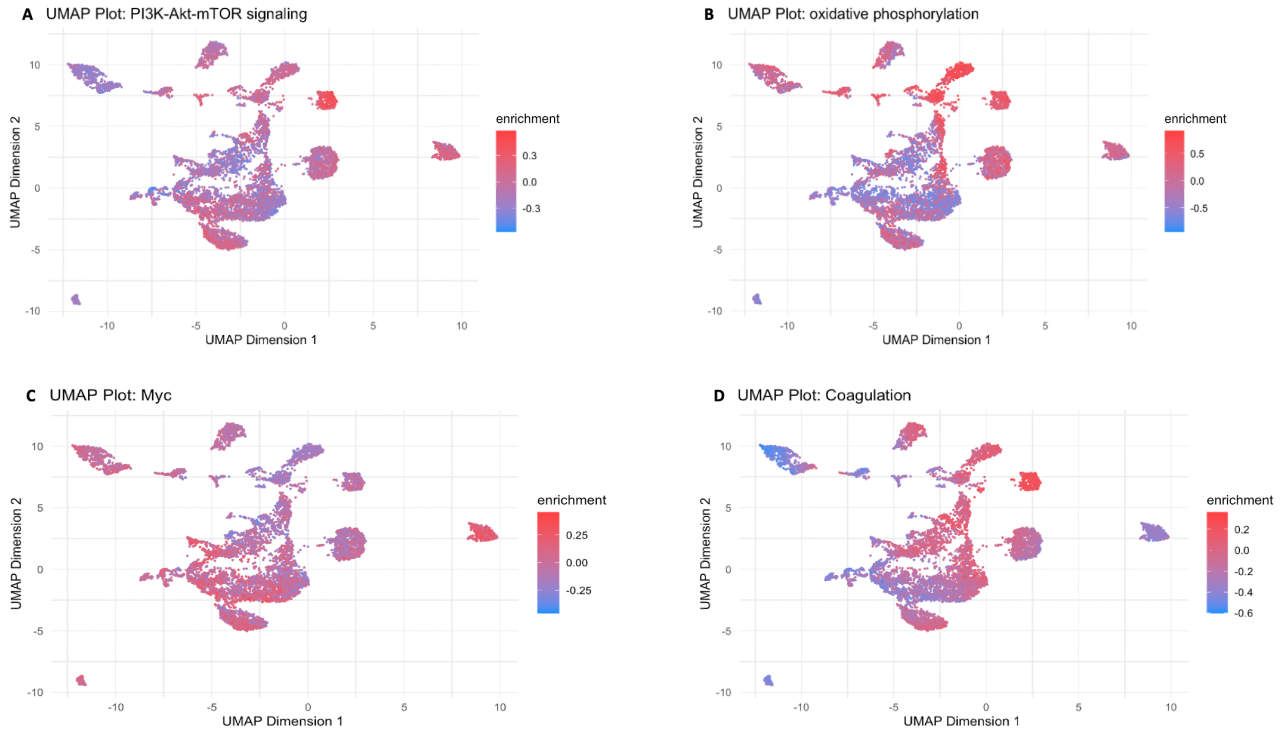

Figure 3.2: UMAP plot colored according to different pathway activities. This figure shows four different UMAP plots of gene expression data colored by their enrichment scores for the activity of the PI3K-Akt-mTOR pathway as well as oxidative phosphorylation, Myc, and coagulation, indicating a tumor-specific enrichment and differences as well as similarities between tumor types.

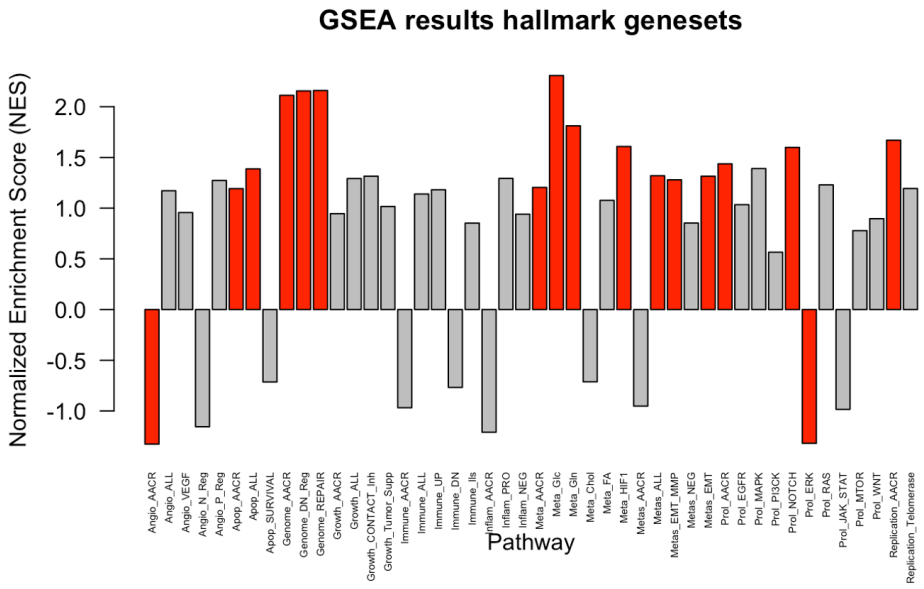

Figure 3.3: Barplot GSEA hallmark genesets. The figure shows the NES value of the given gene sets on the y-lab and the corresponding gene set on the x-lab. Bars shown in red show significant enrichments with a p-value < 0.05.

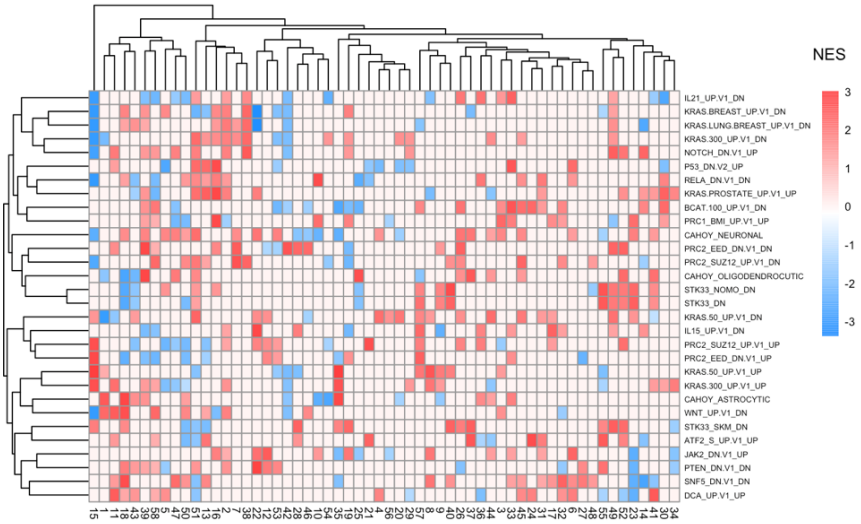

Figure 3.4: Heatmap GSEA patients. The figure shows the the enrichment which each LUAD patient showed in each of the 30 most variant C6 genesets in comparison to healthy tisssue, using a GSEA. Enrichment in the upper part of the ranked list is shown in red colours, enrichment in the lower part is shown in blueish colours. NES values which did not show significance (p-value > 0.15) were set to zero.

3.2.3 LUAD and LUSC patients show differences in gene expression and pathway activity

The comparison of gene expression between LUAD and LUSC patients reveals significant differences. Using a volcano plot, FC is calculated as the ratio of mean gene expression levels between the groups. The Wilcoxon rank-sum test with Bonferroni correction was applied to determine significance (for further detail see methods in the appendix). The results are shown in figure 3.5 A. Next, enrichGO analysis was performed on the 1,500 genes with the highest and 1,500 genes with the lowest FC value. These results can be seen in figure 5.2 in the appendix. Genes associated with epidermal development, keratinization, and cytoskeletal organization are generally downregulated in LUAD compared to LUSC. Conversely, LUAD shows upregulation in pathways related to regulation of body fluid levels, cell-cell adhesion via plasma-membrane adhesion molecules and vascular processes in the circulatory system.

A GSVA was then applied to patients with LUSC or LUAD in order to compare their pathway activity. Again, it is noticeable that both tumor types cluster to a distinct extend, so that a tendency can be identified between LUAD and LUSC (figure 3.5 B).

3.2.4 Binary logistic regression model to differentiate between LUAD and LUSC

The constructed logistic regression model distinguishes between LUAD and LUSC patients based on pathway activity. Following backward selection on an initial model containing the 30 most variable pathways identified through GSVA, 13 pathways emerged as significant contributors (table 5.3 in appendix). Pathways like “HALLMARK_MITOTIC_SPINDLE”, “HALLMARK_OXIDATIVE_PHOSPHORYLATION” and “VEGF_A_UP.V1_DN” indicate higher activity in LUAD. This reflects LUAD’s metabolic activity and interactions with tumor microenvironment.

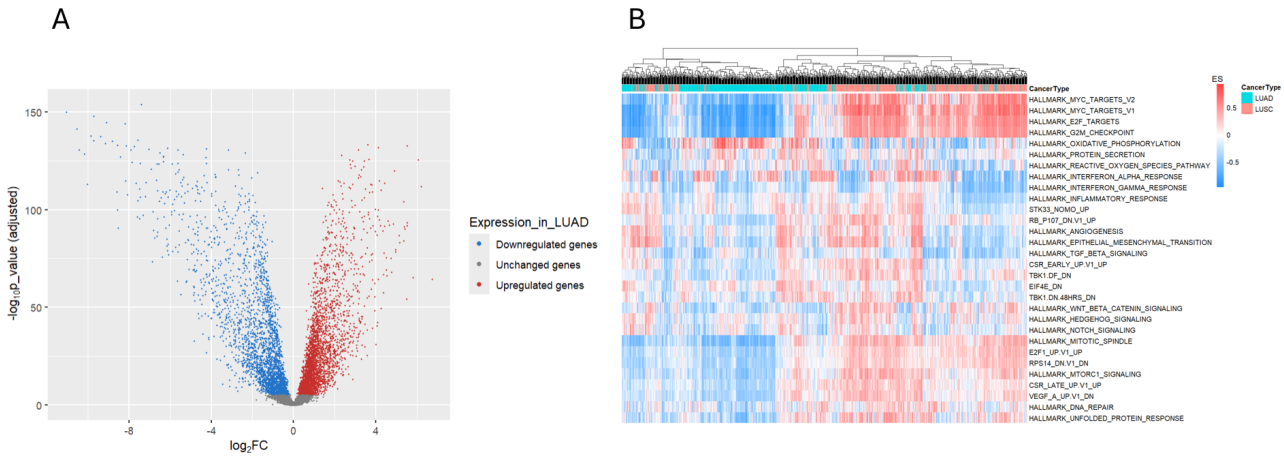

Figure 3.5: Volcano plot and GSVA heatmap LUAD vs. LUSC. (A) This plot visualizes the differences in gene expression patterns in LUAD and LUSC patients. (B) The heatmap displays GSVA enrichment scores for patients with LUAD and LUSC tumor types. It features the 30 gene sets with the highest variance across the rows, while the columns represent individual patients. Color coding at the top indicates the respective classification of patients. Hierarchical clustering groups the patients into two main clusters.

On the other hand, pathways such as "HALLMARK_WNT_BETA_CATENIN_SIGNALING", "TBK1.DF_DN" and "HALLMARK_MTORC1_SIGNALING" are more active in LUSC, as indicated by their negative coefficient. This reflects the structural and keratinozation functions characteristic of squamous cells (see table 5.3 and figure 5.2).

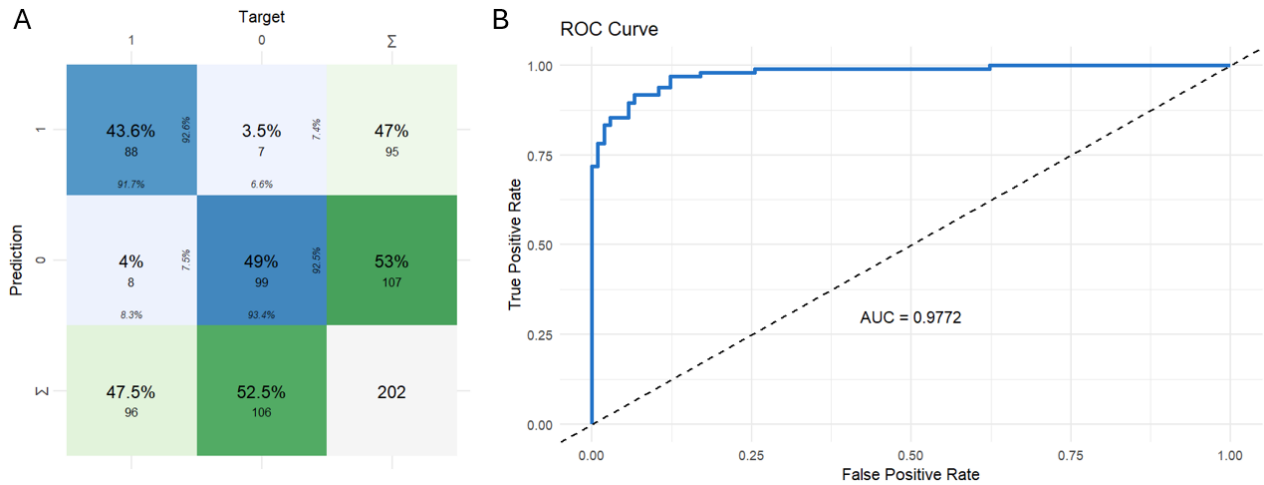

Figure 3.6: Logistic regression model distinguishing LUAD and LUSC. (A) Confusion table summarizing the results of the regression. (B) ROC-Curve visualizing the model's performance.

The model achieved an accuracy of 92.6% in correctly classifying patients, with LUAD identified when $P > 0.5$ on the test dataset. Examination of the confusion matrix reveals accurate classification of 88 out of 96 LUAD and 99 out of 106 LUSC patients, resulting in a false-negative rate of 8.3%, a false-positive rate of 6.6%, and a true-positive rate (recall) of 91.7% (see figure 3.6). Evaluation via ROC curve analysis yielded an AUC value of 0.9779.

Comparison against a null model, assessed using pseudo R-square ($R^2 = 0.7075$) and a chi-square test ($p < 2.2e\text{-}16$), demonstrates significant improvement in predictive accuracy by the logistic regression model. Individual coefficient contributions were evaluated using the Wald test, confirming their statistical significance (see table 5.3).

4 Discussion

4.1 Pan cancer

The development of tumors and cancer is driven by mutations and gene deregulation that affect proliferation, cell death, immune recognition, angiogenesis, and invasion (Poremba *et al.*, 1999). Each tumor exhibits a unique mutational signature influenced by various mutagenesis processes impacting gene expression. Differential enrichment of genesets between tumor types is attributed to genetic diversity, epigenetic alterations, and specific microenvironments (Takeshima and Ushijima, 2019). Genetic heterogeneity also leads to different pathway regulations within a tumor. Additionally, our data include patients with the same primary tumor at different stages, resulting in non-comparable expression levels (Wang *et al.*, 2017). These principles thus validate the results of the UMAP concerning the pan cancer analysis (figure 3.1 B and figure 3.2).
The GSVA results highlight the ratios of up and downregulation in selected gene sets, focusing on the 30 most variant sets for clarity. While more gene sets could be included, the key message remains unchanged, as seen in the hierarchical clustering in the heatmap. Figure 3.5B shows that LUAD has less deviation from 0 on average for the 30 most variant gene sets compared to other tumor types. This suggests that enrichment scores are consistent across the tumor type, supporting a holistic view to identify commonalities for potential uniform treatment approaches. Gene sets describing similar processes in tumor cells, like "HALLMARK_MYC_TARGETS_V2", "HALLMARK_MYC_TARGETS_V1", and "HALLMARK_E2F_TARGETS", which control the cell cycle and proliferation, cluster together (Leone *et al.*, 2001).

4.2 LUAD

GSEA of LUAD patients, comparing healthy and cancerous tissue, reveals key biological processes and pathways involved in cancer progression. Enrichment of hallmark pathways and features such as angiogenesis, proliferation, EMT, and apoptosis suggests common biological drivers of LUAD progression (figure 3.3). However, due to the genetic and epigenetic heterogeneity of cancers, not all hallmark gene sets are enriched uniformly. Individual patient enrichment in specific oncogenic signatures (figure 3.4) reflects the molecular diversity within tumors, with different patients exhibiting varied driver mutations activating pathways like KRAS and PI3K signaling.
The lack of clustering using UMAP (figure 5.1) underscores the variability in LUAD biology, likely due to genetic heterogeneity (TCGA, 2014), microenvironmental regulation (Quail and Joyce, 2013), and epigenetic changes (Jones and Baylin, 2007). These findings suggest that personalized treatment approaches may be necessary for improved therapeutic success, despite common cancer attributes in LUAD.

Comparative gene expression analysis between LUSC and LUAD reveals distinct molecular profiles reflective of their histological origins. LUAD, originating from glandular cells lining the alveoli, shows heightened metabolic and transport activities, while LUSC, arising from squamous cells in the airway epithelium, is enriched in genes associated with structural functions like keratinization (Relli *et al.*, 2019).

4.3 Regression

The binary logistic model constructed to differentiate between LUAD and LUSC patients using 13 selected pathways demonstrates high predictive power. The pseudo R-squared value ($R^2 = 0.7075$) indicates an "excellent model" as it exceeds the threshold of 0.4 (McFadden, 1977). Additionally, the chi-square test results (p < 2.2e10-16) confirm its statistical significance (Wilson and Lorenz, 2015). However, achieving 100% confidence in distinguishing LUAD from LUSC via logistic regression is unlikely, as previous analyses (e.g., GSVA, figure 3.5B) show that LUAD and LUSC share some commonalities and do not cluster completely separately.

To refine predictor selection, a backward selection method was used on a full model initially comprising the 30 most variable pathways identified by GSVA. This method aims to minimize the Akaike Information Criterion (AIC), balancing model complexity and interpretability while avoiding overfitting. However, this approach risks excluding important variables, as once a predictor is removed, it is not reconsidered, potentially overlooking variables significant under different conditions (Chowdhury and Turin, 2020). This limitation could contribute to higher false positive and false negative rates, impacting clinical accuracy. A larger training sample size could improve model validation and generalizability.

The analysis identified key pathways distinguishing LUAD from LUSC, highlighting their distinct molecular profiles. Pathways related to keratinization and cytoskeletal organization were downregulated in LUAD compared to LUSC (figure 5.2), aligning with logistic regression results showing higher activity in LUSC patients for pathways like "HALLMARK_MTORC1_SIGNALING" and "HALLMARK_WNT_BETA_CATENIN_SIGNALING" (see table 5.3) (Relli *et al.*, 2019). Conversely, "HALLMARK_OXIDATIVE_PHOSPHORYLATION" and "VEGF_A_UP.V1_DN" were positively associated with LUAD, with increased oxidative phosphorylation suggesting enhanced metabolic activity crucial for meeting tumor cells' energy demands and supporting rapid growth.

Understanding these molecular distinctions offers insights into tailored therapeutic strategies. Targeting specific dysregulated pathways, such as blocking mitochondrial oxidative phosphorylation, could potentially enhance treatment efficacy for LUAD patients (Fulda *et al.*, 2010) (Wang *et al.*, 2024).

4.4 Outlook

Our analyses on LUAD provides a foundation for future research aimed at deeper insights and clinical applications. Moving forward, it is essential to explore the genetic heterogeneity within LUAD tumors. Understanding these variations could reveal their impact on disease progression and treatment response. Integrating other data on e.g. DNA methylation, proteomics, and metabolomics also holds promise for

comprehensive molecular profiling of LUAD, and might contribute to the identification of novel targets in treatment and diagnostics of LUAD. Also, a validation of our findings using independent datasets and experimental models is crucial. Translating identified biomarkers and pathways into clinical assays could lead to diagnostic innovations and tailored treatment approaches in LUAD patients especially when it comes to differentiating between LUAD and LUSC and hence optimizing a patient's treatment. Harnessing biomedical data science, machine learning, and artificial intelligence could play a pivotal role in developing robust algorithms for early detection, prognosis, and therapeutic response prediction in LUAD. Our findings may contribute in further developement of these advanced tools.
In conclusion, our findings underscore the complexity of LUAD and highlight opportunities for further exploration and innovation in cancer research. Further work in these future directions of cancer research might improve outcomes for LUAD- and other cancer-patients through both personalized and targeted therapeutic approaches and advanced novel diagnostic methods.

5 Appendix

5.1 Methods

5.1.1 Volcano plot

A Volcano plot is a valuable method for comparing gene expression differences between two groups. Initially, the fold change (FC) value is computed using the formula:

$$\text{FC} = \frac{\text{mean(group 1)}}{\text{mean(group 2)}}$$

Next, a Shapiro-Wilk test assesses the normality of the data distribution, guiding the selecting of appropriate statistical tests to determine significance (Yap and Sim, 2011). Given the non-normal distribution indicated by the Shapiro-Wilk test, the Wilcoxon rank-sum test is employed for further analysis. To account for multiple comparisons, the Bonferroni correction is applied to the p-values. This approach allows the visualization of gene expression differences between LUAD and LUSC patients, with LUAD as the first group and LUSC as the second. Gene expression data from the pancancer data set was utilized.

5.1.2 Detailed Explanation of Binary Logistic Regression

As already explained, the logistic regression model is represented as:

$$\log\left(\frac{p_1}{1-p_1}\right) = \beta_0 + \beta_1 X_1 + ... + \beta_i X_i$$

where: p is the probability of the event, β_0 is the intercept, β_1, β_i are the coefficients for the predictors X_1, X_i. These results are tranformed by the logit function $\log(\text{odds}) = \log\left(\frac{p_1}{1-p_1}\right)$ ranging from $-\infty$ to $+\infty$. A regression line is fitted to these data points, representing the relationship between the logistic probability and the predictors. The initial data points are projected onto this line to determine

coefficients and a new logit value is calculated. The sigmoidal curve is plotted by applying the logistic function to the new logit value, thereby obtaining a value between 1 and 0:

$$p = \frac{e^{\log(\text{odds})}}{1 + e^{\log(\text{odds})}}.$$

The best-fitting regression line is identified using the "maximum likelihood" method, which calculates probabilities for each data point. To determine the optimal line, likelihoods are computed for all possible lines with slight adjustments to their parameters in each iteration. The line achieving the highest likelihood value is considered the best fit (Wilson and Lorenz, 2015).

5.2 Plots

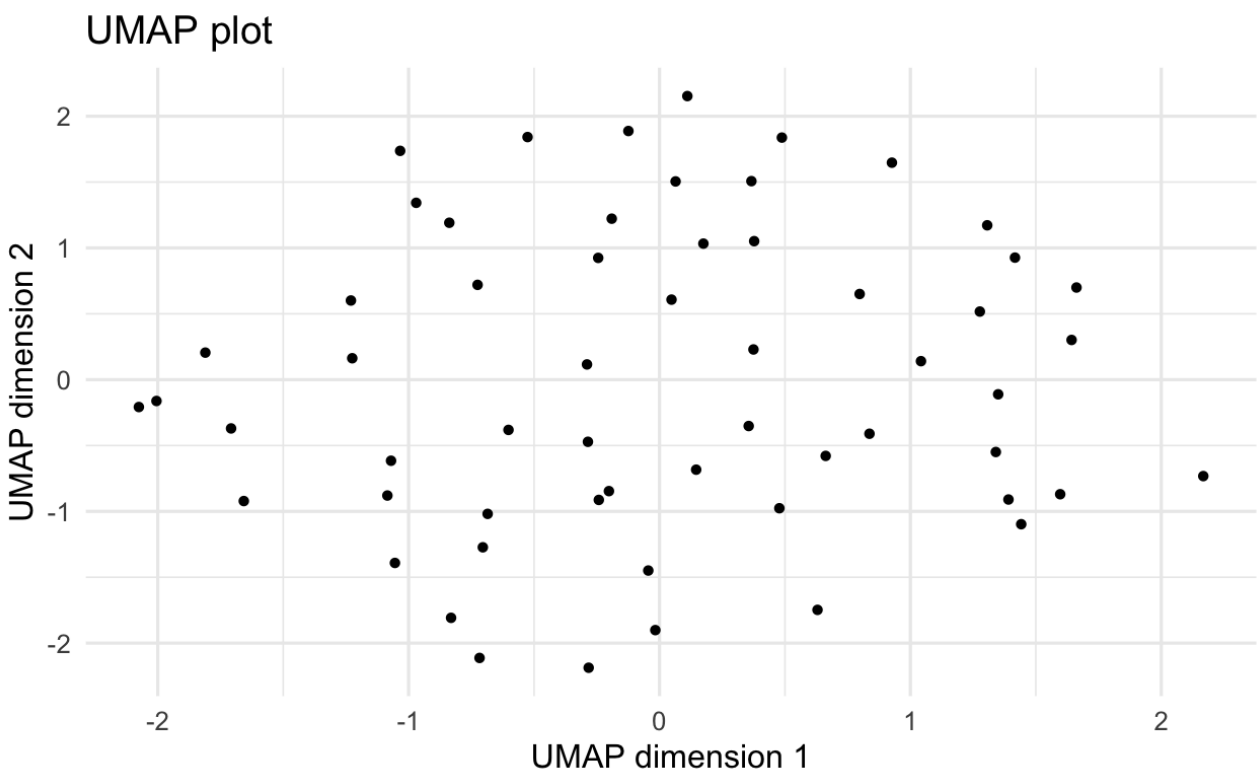

Figure 5.1: UMAP GSEA patients. This figure shows the clustering of LUAD-patients using the GSEA data on foldchange values between healthy and cancerous tissues using both hallmark and oncogenic signature genesets, via UMAP.

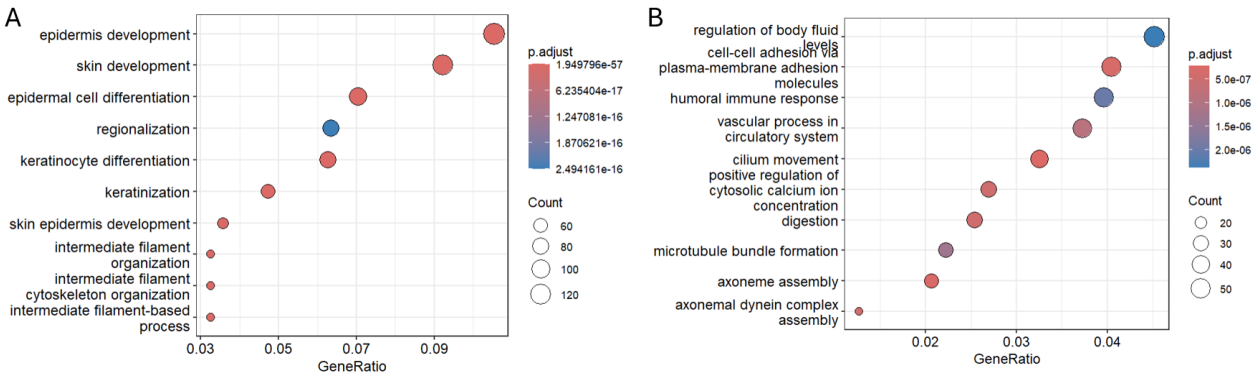

Figure 5.2: enrichGo analysis of 1,500 highest and 1,500 lowest FC values of comparisson of LUAD and LUSC. (A) This shows processes associated with the 1,500 gene with lowest FC value. (B) Here, upregulated processes associated with the 1,500 gene with highest FC value can be seen.

5.3 Binary logistic regression

Table logistic regression model variables and stats

	Estimate	**Standard error**	**p-value**
Intercept	-0.1503	0.1785	0.399783
HALLMARK_MITOTIC_SPINDLE	4.2191	1.1547	0.000258
HALLMARK_MTORC1_SIGNALING	-4.4199	0.9632	4.46e-06
HALLMARK_NOTCH_SIGNALING	1.9501	0.7373	0.008176
HALLMARK_OXIDATIVE_PHOSPHORYLATION	2.1717	0.4644	2.92e-06
HALLMARK_PROTEIN_SECRETION	2.6732	0.7092	0.000164
HALLMARK_WNT_BETA_CATENIN_SIGNALING	-4.5210	0.7047	1.40e-10
E2F1_UP.V1_U	-5.1325	1.4770	0.000511
EIF4E_DN	3.8867	1.0141	0.000127
RPS14_DN.V1_DN	-5.5243	1.5983	0.000547
STK33_NOMO_UP	-4.0512	1.0705	0.000154
TBK1.DF_DN	-5.5171	1.0151	5.47e-08
TBK1.DN.48HRS_DN	2.8749	0.7334	8.85e-05
VEGF_A_UP.V1_DN	4.1303	1.4051	0.003288

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